Speaking Their Language

An LLM-First Approach to Persistent Characters in Near-Natural Worlds

Bruce D'Ambrosio
Oregon State University

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1 Abstract

Just as one justification for humanoid robots is they must function in spaces designed for humans, useful AI must co-exist with us in natural and near-natural environments (NNEs). We argue that NNEs are quite different from artificial ones like coding or synthetic 'planning' domains, are understudied, and present a unique opportunity for large language models (LLM)s. We present a text-first architecture in which LLMs are the *sole* problem-solving element for persistent, situated characters in NNEs. The central hypothesis is that LLMs not only possess an innate ontology of these worlds, enabling abilities to interpret, infer, and predict, but also an embedded phronesis[1] for acting competently therein. Rather than coercing behavior via finetuning, we scaffold NNEs with a closed-world *Ground* (primitive nouns/verbs) and a scenario-aware *Language* layer (ontology, activity policies, plan grammar) so the model can plan and act using its native competence. At runtime a distributed OODA loop (Observe-Orient-Decide-Act) coordinates world simulation, activity selection, plan generation with control flow and social memory. Early domain experiments (e.g., (1) a rural village setting inspired by *La Terre* and (2) a lightweight forest survival setting) show coherent multi-turn behavior, interpretable plans, and measurable drive fulfillment with minimal non-linguistic machinery.

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2 Motivation and Thesis

Much recent research has focused on getting LLMs to exhibit longer-term multi-turn coherence. Both external methods, *e.g.* early attempts at chatbots (*ie*, recycle previous input-output into context), RAG, on-line ICL, *etc*. and internal "policy-in-weights" approaches (SFT/RLHF/DPO/GRPO) can be seen in this light. Perhaps less attention has been paid to studying LLMs as language producing/consuming artifacts. LLMs produce streams of tokens that, run through a tokenizer, *look* like words we recognize, sequenced in a way that again seems recognizable.

We suggest this apparent recognition is misleading. LLMs have been trained on a large corpus of text spanning time, space, and language. Words, and even worldviews, vary in meaning depending on context. One strong indicator of this is the classic prompt guideline to start a prompt with 'You are an expert …'. Perhaps we would do well to follow the methods of anthropologists or linguists when faced

1. a Greek term often translated as "practical wisdom" or "moral intelligence," referring to the ability to make morally good decisions in practical situations. It involves understanding the right course of action based on experience and ethical reasoning. A less ethically loaded term might be *techne*, technical or practical skill.

with an unknown language: slowly build an ontological mapping of nouns and verbs through careful discourse with the native-speaker while referencing objects and actions in the environment.

Main claim: if we speak the LLM's language—NNE-appropriate nouns/verbs, activities and social expectations grounded in the environment, an LLM can sustain coherent behavior in NNEs without algorithmic glue beyond light scaffolding. This differs from both early attempts at chatbots (*ie*, recycle previous input-output into context) and from more recent post-training "policy-in-weights" approaches (SFT/RLHF/DPO/GRPO) to eliciting coherent longer-range behavior. We derive though conversation a textual representation of the innate *weltanschauung* (ontology, goals, plans) of an LLM, and let the model deliberate in text while we strictly control the symbol surface it can use to act.

Conjecture: NNEs are quite different from artificial ones like coding or synthetic 'planning' domains. In particular, LLMs have capabilities in these environments beyond those seen in testing in artifical domains, are understudied, and present a unique opportunity for large language models (LLM)s. Further, the difference between a *programmer* and an *AI coding agent* is largely accounted for by the ability of the former to operate flexibly in an NNE, the development group.

Conjecture: LLMs encode rich priors about NNEs from pretraining.
LLMs, through their pretraining on vast corpora of human-generated text (often drawn from the internet, books, and other sources reflecting everyday life, stories, and societal interactions in great detail), develop an implicit ontology of (NNE) worlds , alongside a probabilistic epistemology where knowledge manifests more as predictive accuracies than as justified true beliefs, shaped by inductive biases that favor default interpretations. The specification of a setting induces not only an ontology, but also a *phronesis*, or at least a *techne*. These *weltanschauung*[2] are highly context sensitive – for the same LLM, a setting of a modern trading floor in Manhattan will queue a very different ontology and phronesis than that of an 18th century peasant farm.

Conjecture: NNEs should be "Turing-compatible"
Here we refer not to the LLM, but to the environment in which it is expected to operate. Our plan is to ground the LLM-derived ontology in the primitives provided by the environment API. We suggest that one guideline for this API is that it be Turing-compatible, that is, more-or-less Turing-complete, absent infinities. This yields open-ended behavior (*ie* anything achievable via computation) with a small action set, presenting interesting 'planning' challenges.

Conjecture: The right language - closed-world primitives + ontology + activities - lets the LLM act reliably without heavy non-linguistic algorithms.
Another way to say this is that LLMs capture not just ontology, but *behavior.* Further, the encoded behavior is sufficient for an LLM to act robustly, *if the innate and external ontologies are aligned and mostly co-extensive.*
Alignment refers to the sense of a word. *Inheritance*, for example, might reside in a quite different context 19th century rural peasant France (*eg* social norms or community expectations) than in 21st

2 We will subsequently refer to the *phronesis* simply as 'activities'.

century urban culture (*eg* will, revocable trust, …). By co-extensive we refer to the coverage of the ontology and activities. Informally: (1) almost all the objects the LLM 'expects' to exist in the environment are mappable into ground nouns, and (2) that almost all the activities the LLM 'expects' to be able to perform in the environment are mappable into sequences of API actions.

Observation.

Initial LLM-mediated offline analysis can reveal serious incompatibilities between an LLM innate model of a environment and the world presented to it in the API. These incompatibilities can occur both in alignment and coverage.

3 Architecture / First Implementation

We distinguish *Setting* (era, locale, climate, drives, social norms, character descriptions), which queues a *weltanshauung*, from *Ground*—the primitive, closed-world ontolog**y** of nouns and verbs available in the API through which the LLM interacts with its environment. The ground pins the LLM's inputs and outputs to finite sense and action vocabularies. In our experiments to date, the environment is a grid-map 2½ D simulation with terrain, properties/infrastructure, resource instances, characters, together with a simple sense / act API. The API is, we believe, Turing-compatible – it includes the equivalent of move, read, write, and conditional operations. It seems sufficient to model the world of physical action. One lack of co-extensivity recently discovered is that the world has no primitives corresponding to objects of *discourse:* agreement, position, and other cognitive artifacts of human discourse. We intend that this be one focus of future work.

3.1 LaTerre

LaTerre, by Emile Zola, is a novel about mid 19th century peasant life in rural France. Praised for its unflinching realism, one focus is the tension between father and son over land inheritance.

```
Setting
Summer morning in rural France, 19th century, a small village in the
countryside. The village is a small farming community with a few farms
and a church. The village is located in a valley with a river running
through it.

Characters
Jean Macquart, a hardworking young unmarried peasant farmer working his
father's farm. You left military service to return to the family farm.
You are strong, honest and committed to working the land, but have a
quick temper. You speak plainly and directly, in the style of a volatile
19th century french peasant speaking to an acquaintance. You hope to
inherit a share of the family farm and make a living as a farmer. Despite
being french, you speak in peasant-french accented english. Your name is
Jean. FatherMarquart is your father. He has not yet died, but is old and
ill, and has not yet deeded the farm to you.
drives
i gain your rightful inheritance - justice and fairness in how the land
is divided
```

```
ii maintain and work the family farm
iii find love and a wife to build a family with
iv immediate needs of survival - food, shelter, health, rest from
backbreaking labor
```

3.2 Language as Control Surface

3.2.1 Ontology

Between free text and ground sits an **ontology**: two DAGs (verbs, nouns) with 'refine' edges. These specialize abstract activities ("Acquire", "Cultivate") and categories ("Food", "WaterSource") toward the ground (*eg* 'take', 'use'; and 'egg', 'well' respectively). The ontology is setting-aware: it is defined through interaction with an LLM when a character enters a setting, ensuring activity/goal phrasing stays within the closed world while still leveraging the model's richer conceptual space.

3.2.2 Phronesis

Phronesis, knowing how to act ethically (or *techne*, roughly, practical action), emphasizes knowledge and action. A radical reader might say this is a rejection of planning. We don't *plan*, we *know.* In concert with ontology we ask the LLM to describe a set of activities that cover activities of daily living for the character in question in the given environment, focusing especially on the given drives, and using the ontology (the process we use is a two-stage co-evolution of ontology and phronesis). Interestingly, in the LaTerre world, for Jean, the LLM identified a spatio-temporal range of activities, from having breakfast at the farmhouse to fall harvest. This suggests the view that breakfast isn't a subgoal of the larger fall-harvest project, but rather an independent situated activity temporarily preempting harvest.

3.3 Templates (Specifications)

As described earlier, we begin by eliciting an ontology and set of activities through a two-stage dialog. For this and other tasks we use prompt templates that describe the task and response syntax. In initial offline analysis:

- **Setting/Drive analysis**: Given a setting, character, and a specific drive of that character, infer a small set of **axes** (e.g., *Thirsty* → *Hydrated*, *Lost* → *Oriented*) and propose Activity–ItemClass pairs (verb-noun pairs) that move character state along the axis. Under a closed-world constraint, each ItemClass noun is mapped to included/excluded primitives. This is iterated over all character *drives*, exposing the model's latent ontology for this character in this setting. It also binds this ontology stably to the environment API. For example, one drive of Francoise, a young unmarried woman in *LaTerre*, is to protect her reputation. One axis the LLM identified for this was *At-Risk-of-Scandal*, and one activity-itemclass pair generated for this axis was to attend public gatherings in respectable places. The complete activity-itemclass set for this drive is in an appendix:

```
{"axis": ["At-Risk-of-Scandal","Not-At-Risk-of-Scandal"],
"drive": "avoid scandal and protecting your reputation",
```

```
"activity_itemclass": {"activity": "Attend", "itemclass": "Public-Gathering",
"subtypes_included": ["Church","Market"]}}
```

- **Activities generator**: Produce 24–40 scenario-appropriate Activities of Daily Living (ADLs) and instrumental activities with schematic steps expressed **only** in allowed nouns/verbs. Activities serve as action option templates with 3–5 schematic steps, e.g.:
    - *Drink from Well* (daily@morning): move to Well; take Water; drink Water.
    - *Plan Day's Fieldwork*: think about Field; inspect Tool; place priorities.
    - *Rest after Fieldwork*: move to Farmhouse; rest in Farmhouse; recover in Farmhouse.

  Because steps are expressed **only in allowed nouns/verbs**, the rewrite/goal→plan path (below) remains grounded. Activities also tag **states addressed** (e.g., "thirst", "fatigue"), letting the *Orient* phase at runtime trade off drives against context. Activity selection is performed primarily by the LLM, following simple attribute-level (*e.,g.* Time and place) filtering. The activity generator will occasionally add nouns or verbs not identified in Setting/Drive analysis (*eg*, a 'milk the cow' activity despite the lack of cows in the API or the ontology). We view this as a feature, not a bug: it reveals an incompleteness of the provided environment wrt the LLM's expectations for the specified setting. Another drive of Francoise is to help with family chores. An example activity:

```
"FetchEggsFromChickenCoop": {
"name": "FetchEggsFromChickenCoop",
"category": "Instrumental ADLs/Logistics",
"tags": ["physical","routine","outdoors","solo"],
"steps": ["move to Chicken","obtain Egg","return to Farmhouse"],
"when": "daily@morning",
"where": ["Chicken"],
"duration": [10,15],
"needs": ["Chicken"],
"states_addressed": [],
"importance": 0.6,
"habit": 0.85
}
```

Then, at runtime:

- **Activity selection:** Select an activity from the set of activities generated offline. (Fallback to generating an on-the-fly goal if no activity seems appropriate)

- **Goal template** and **Step-rewrite template**: Turn either a free goal or the next activity step into a precise situated goal ("Reduce thirst" → "Drink from Well: … termination: Thirst < 50").

- **Plan generation**: Synthesize an explicit plan from the plan grammar with steps and control flow (if, while, scan out var), using only the primitive verbs and bound targets. The grammar is

intentionally small and human-auditable. There is planning, but only for small, local problems, and plans are LLM-generated.

A plan is accepted only if it validates against the grammar; otherwise the system reproposes.

A counter-argument to our claim that the LLM is doing all the problem-solving is that this flow is itself the core ‘algorithm’ of our system, together with its run-time architecture. Nonetheless, This policy-in-text approach yields interpretable, verifiable outputs from each stage of reasoning while preserving the LLM’s freedom to reason in natural language inside the templates. None of the templates specify *methods* to produce the desired output, they merely specify *what* is needed and the *format*.

3.4 Character Run-time Loop (OODA)

3.4.1 Observe

Each character receives a compact situation summary: current location/terrain/property, visible resources/persons with distances, adjacency affordances (nearby interactables), inventory, last actions, and entity-centric dialog memories. Observations are formatted as text and fed into the templates.

3.4.2 Orient

Orientation follows a priority ladder:

1. *Physiological override* if any tracked state (hunger, thirst, fatigue) crosses critical thresholds → generate a remediation goal.
2. Otherwise, if a *plan* is active, execute the next plan-step.
3. Otherwise, if an *activity* is active, rewrite the next activity-step into a concrete goal (speaking in the closed-world vocabulary), generate a new plan, and execute its first plan-step.
4. Otherwise, try to select an appropriate activity and begin executing it.

5. Otherwise, propose an **ad-hoc goal** from context and plan and begin executing on it.

3.4.3 Decide (Plan)

Given the current goal and observations, the planner emits a JSON plan using a small action description language:

- **Primitive actions**: *move(target), take(target), place(target), inspect(target), use(target), scan(target-class, out_var), say(target, value), think(value).*
- **Control flow**: *if <condition>, while <condition>*. Conditionals include *bound(target_var), near(target), canSee(target), and has(target).*
- **Binding**: *scan* binds *out_var* (e.g., $nearest_well) to the nearest instance. All other primitives can reference variables and/or instances.

3.4.4 Act

The executor resolves targets (e.g., $nearest_well), checks preconditions (visibility, distance, inventory), and applies outcomes (state updates, inventory changes, world mutations). Actions are logged with timestamps, distances, and post-state snapshots. Multi-agent concurrency uses simple **conversation locks** for social acts and a **turn manager** advances **simulation time** based on per-character proposals (the shortest-duration proposal is accepted for advancing the clock.).
Some notes on primitive actions:

- *Scan* accepts any ontology noun and binds to the nearest visible instance.
- *If* and *While* have the usual python semantics
- *Wait* passes on acting until its condition becomes true (or timeout).
- *Take* and *Place* provide Turing-machine-like ability to remove and write symbols into grid cells
- *Inspect* provides something like a Turing-machine ability to read a cell, *except* that the response is context-dependent. 'Attributes' of objects are not fixed, either by object class or setting, but rather determined dynamically in context.

On plan completion, a reflective summary of plan execution is created, both as instrumentation and for later reflection.

3.5 Character Cognitive Architecture

We have only a minimal cognitive architecture at present. The following list both records current implementation and implies (in the 'without' clauses) future plans (mostly drawn from prior versions of the platform yet to be ported). In all cases below, very limited situated-query capabilities exist:

1. Sense data is recorded as received, without an filtering, prioritization, or transformation into perceptions (drive-relevant statements).
2. Conversations are recorded per-character and summarized, but without any explicit formation of TOM models of others. This enables minimally coherent conversations between characters, but does not include any explicit recognition of individual or shared commitments made in conversation. Conversation occurs 'offline', that is, outside the planning loop. A major focus of current work at this time is to extend the system to integrate discourse into the ontology and architecture.

3. Consequences of *inspect* and *use* acts are recorded, but without any explicit models formed of individual objects or object classes.
4. Episodic memory is implemented as a simple action history plus reflective summarization of each plan execution, without embedding under time, place, goal, or any other dimension of mental state.
5. Spatial memory is absent. 'Visual' input does support SLAM, but this hasn't been integrated.

3.6 Measurement and Instrumentation

We log per-plan outcomes and maintain detailed **telemetry**: observable and cognitive-state histories, step success/failure reports, bound-variable resolutions, per-step **physiology** values, and proposed **time advances**. Plan-level summarizations include:

- **Goal satisfaction** (plan-local termination fulfillment).
- **Drive fulfillment/alignment** (aggregated change on tracked axes).
- **Scenario score** combining recent drive fulfillment and goal succes.

These are intended for within-subject ablations (e.g., baseline vs. baseline + derived upper-level ontology; + ADLs). The focus is **comparative**, not absolute scoring.

4 Experiments

We use two of our lightweight NNEs in the experiments reported here:

1. **Rural village (19th-century)** inspired by *La Terre (Emile Zola, 1887): (*places (Farmhouse, Field, Market, Church, Well, Mill), tools (Hoe, Scythe, Plow), foods (Bread, Cheese, Milk, Eggs), people, and land/inheritance themes. Drives include "work the farm", "secure fair inheritance", "form a family". Activities cover daily farm routines, social/legal interactions (inspect deed, discuss inheritance), and production chains (scythe → hay; milk cow → milk).
2. **Forest survival / egress ('Lost')**: basic hydration/energy/alertness, movement through forest/clearing/outcrop, landmarks, and safety checks; activities like "Enter Clearing and scan", "Navigate to edge", "Drink at spring".

4.1 Findings (qualitative)

- The **Language layer** (ontology + policy templates) reliably constrains plans to valid primitives while preserving flexibility.
- **Scan→bind→act** patterns are frequent and useful (e.g., scan(WaterSource, $nearest_water); move($nearest_water); use($nearest_water)).
- ADLs give robust baselines; physiological overrides interleave sensibly with role/production tasks.
- The system is capable of generating natural but surprising plans, ones that we believe might be unlikely to occur to builders of traditional abstract planners. For example (paraphrased): I'm tired, look for a shelter. If I don't see one, look for a thicket (to hide in). *If I don't find one, wait for Joe and ask him if he knows where there is a shelter.*
- Despite the lack of a discourse ontology, social exchanges stay coherent across turns due to entity-centric dialog memory.

4.2 Findings (quantitative)

First we instrumented to measure a number of basic performance metrics with and without ontology and activities. The without case generates fallback ungrounded goals (in free text, without guidance from the ontology), and can only scan for primitive nouns.

avg_plan_score	6	0.8631	0.0912	6	0.7751	0.0661	0.0880	0.0460	1.9140	0.08748
avg_state_mean	6	0.0338	0.0127	6	0.0641	0.0254	-0.0304	0.0116	-2.6205	0.03296
avg_failure_rate	6	0.0120	0.0125	6	0.1297	0.0938	-0.1177	0.0386	-3.0473	0.02726

- **avg_state_mean**: *increase* in average hunger / thirst / fatigue level
  - Good – baseline was pursuing physiological goals too often, as indicated by a low state mean of 0.0338 (0.0 = 0 physiological need, 1.0 = death)
- **avg_failure_rate**: *increase* in avg plan failure rate.
  - Unclear interpretation – more scan fails? Tbd
- No other clear differences.
  - Expected – these are *local* per-plan measures. We haven't made things worse locally.

Longer-term metrics

In this second experiment we evaluated a total of 7 characters in 3 scenarios over 100 turns each. The key measure is momentum. The momentum template is shown below, and attempts to evaluate whether the character shows progress over the 100 turns, as opposed to merely performs repetitive actions that align with the situation and character drives.

As shown, the difference between no activities and no ontology (FF) vs with both activities and ontology (TT) is significant using Welch t-test at .5%, and using Mann-Whitney U at .2%

AO	Coherence	Alignment	Realism	Momentum
FF	5.7	6.04	5.83	3.58
FT	5.625	6.4	6	4.17
TT	5.75	6.375	5.92	5.42

5 Discussion

5.1 Why This Works (and What It Is Not)

- **Works** because the model operates in a **language it already knows**: NNEs with culturally typical nouns/verbs, social scripts, and daily activities. We provide a crisp **control surface** (closed world + plan grammar) so outputs are usable by a simulator and comprehensible by humans.
- **Not** an "agent framework" that fills behavior with procedural code. Our non-linguistic layer is intentionally thin: map mechanics, inventory, turn/time, and validators.

5.2 Limitations and Open Problems

- **Ontology leakage** and over-abstract verbs/nouns if the middle layer is too thin; too thick makes it brittle. We counter by (i) bounding depth and (ii) validating all steps against Ground.
- **Plan fragility** around target resolution and distance checks; add richer precondition tests and better scan heuristics.
- **Resource lifecycles** (production/consumption, dynamic types) are largely ignored.
- **Multi-party acts** (e.g., negotiation) require better locks/turn-taking and richer speech acts ("invite", "promise", "agree", "request-info").
- **Evaluation** still relies on proxy metrics; we are building more evidence-linked scenarios and scoring, along with reflection/critic loops (plan problems with severity, schema violations, efficiency issues).

5.3 Contributions

1. **Language-first control surface** for NNWs: closed-world primitives, middle ontology, and policy templates—**policies as text**.
2. **Explicit plan grammar** with control flow and variable binding that stays within the LLM's comfort zone while being simulatable.
3. **Persistent characters** with lightweight OODA, social memory, and activity options (ADLs) that yield coherent, ongoing behavior with little non-linguistic logic.
4. **Instrumentation & metrics** suitable for ablation studies on language/ontology components rather than finetuned weights.

6 Roadmap

- Expand speech-act set and conversation norms; enrich **entity models** with roles, trust, and commitments.
- Formalize **dynamic resource production/consumption** and inventory semantics.
- Harden **validator/runner** (preconditions, retries, recovery goals).
- Broaden **activity catalogs** per setting and automate **middle-ontology** checks.
- Tighten **reflection pipelines** to improve prompts/templates from evidence-linked plan logs.

7 Conclusion

In NNEs, the shortest path to persistent, coherent behavior is to **speak the model's language**. A closed world of primitives, a thin but expressive middle ontology, and text-specified policies allow LLMs to plan and act with minimal extra machinery. The result is a system that remains interpretable, data-driven, and extensible—ready for systematic ablations on what language structures matter most for grounded competence.

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Language-to-action for long-horizon household tasks; standard for evaluating instruction following in homes.

Appendix 1: Example (partial) initial Ontology


```
{"axis": ["At-Risk-of-Scandal","Not-At-Risk-of-Scandal"],
"drive": "avoid scandal and protecting your reputation",
"activity_itemclass": {
"activity": "Secure",
"itemclass": "Legal-Document",
"subtypes_included": ["Will","Deed"]}},

{"axis": ["At-Risk-of-Scandal","Not-At-Risk-of-Scandal"],
"drive": "avoid scandal and protecting your reputation",
"activity_itemclass": {
"activity": "Attend",
"itemclass": "Public-Gathering",
"subtypes_included": ["Church","Market"]}},

{"axis": ["Unknown-Intentions","Known-Good-Intentions"],
"drive": "avoid scandal and protecting your reputation",
"activity_itemclass": {
"activity": "Associate-With",
"itemclass": "Respected-Person",
"subtypes_included": ["FatherMarquart","MotherMarquart"]}},

{"axis": ["Unknown-Intentions","Known-Good-Intentions"],
"drive": "avoid scandal and protecting your reputation",
"activity_itemclass": {
"activity": "Use",
"itemclass": "Public-Institution",
"subtypes_included": ["Church"]}},

{"axis": ["Isolation","Community-Integration"],
"drive": "avoid scandal and protecting your reputation",
"activity_itemclass": {
"activity": "Visit",
"itemclass": "Community-Place",
"subtypes_included": ["Church","Market","Village","Farmhouse"]}},
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